Implementation of a Multi-Modal Predictive Model for Cloud Cyber Attack Detection Using CNN-LSTM Architectures

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***Abstract*—This paper emphasizes the implementation of multi-modal Intrusion Detection System architecture for state-of-the-art cloud environments. While cloud infrastructures are becoming large and complex, it is difficult for traditional security mechanisms to provide protection across distributed resources comprehensively. For this reason, the proposed system brings together telemetry from different layers, such as hypervisor logs, guest virtual machine performance metrics, and network flow data. Combining such diverse data sources expands the model's view to become more broad with context about system behavior.**

A hybrid CNN-LSTM architecture forms the core of the system, wherein the convolutional component identifies meaningful spatial patterns in the structured telemetry data, while the LSTM component captures temporal relationships common in multi-stage attacks such as Distributed Denial of Service and brute force attempts. A synthetic data pipeline was developed to simulate realistic attack behavior, enabling controlled experimentation and validation of the detection performance. Along with model design, this work explores containerized deployment using cloud-native orchestration platforms and discusses federated learning for multi-tenant cloud security environments as a means of enabling privacy-preserving knowledge sharing among tenants.

***Index Terms*—Cloud Security, Deep Learning, CNN-LSTM, ISOT-CID, MLOps, Containerization.**

I. INTRODUCTION

Cloud computing technology is the back end of the information age. It supports most of the critical infrastructure and services around the world, including financial services, healthcare systems, the systems that deliver online commerce, and even big data analytics. With the inclusion of even more advanced cloud computing technologies like containers, microservices, and serverless computing, the observed scalability is matched by the growing attack surface for cybercriminals. These solution processes also gradually lose their effec-tiveness over time. For cloud ecosystems, these processes are based on rigid rules, signature-based systems, and a rigid defensive posture. Cloud infrastructures have a dynamic and multi-tenant nature. Resources will be dynami-cally provisioned and decommissioned, with workloads being migrated and communication happening across the multi-layer structure of virtual machines, hypervisors, and more. All these characteristics make it difficult for a traditional intrusion detection system to provide visibility in a timely manner.

In recent times, several reports from various threat intel-ligence reports indicate that hackers are now using identity-based attacks, large-scale distributed denial of service attacks, and highly sophisticated advanced persistent threats.

In addi-tion to this,hackers are using artificial intelligence tools for carrying out automatic phishing attacks, creating malicious codes, and evading detection systems. A new dimension of security risk lies in machine learning pipelines that are being threatened by data poisoning and machine learning evasion.

To overcome such challenges, predictive intrusion detection techniques based on machine learning and deep learning techniques have seen increasing interest in addressing the security challenges in the cloud environment using cloud telemetry data. The techniques such as Random Forest, Support Vector Machines, and hybrid models such as CNN-LSTM have shown promising results in effectively addressing various types of complex intrusion patterns in cloud telemetry data.

This paper proposes a cloud-native intrusion detection system framework with synthetic data simulation, feature extraction, machine learning baseline models, deep learning models, and a cloud-native architecture based on containerization platforms. With these components, the framework supports intrusion detection with the benefits of scalability, interpretability, and flexibility as envisioned for contemporary cloud-based architectures.

The proposed architecture supports an end-to-end pipeline that includes data generation, feature engineering, training, experiment tracking, and deployment as an API service. It has the ability to leverage deep learning methods in addition to traditional techniques, along with appropriate deployment methods in the cloud, in order to address contemporary cloud security challenges in a most effective way.

II. RELATED WORK

The concept of cloud security changed dramatically with the increase of distributed and virtual computer environments.However, with the move to cloud computing environments, there are new challenges such as multi-tenancy, dynamic resource allocation, and distributed data generation at a massive scale, which demand intelligent intrusion detection techniques that are efficient in analyzing heterogeneous cloud environments.

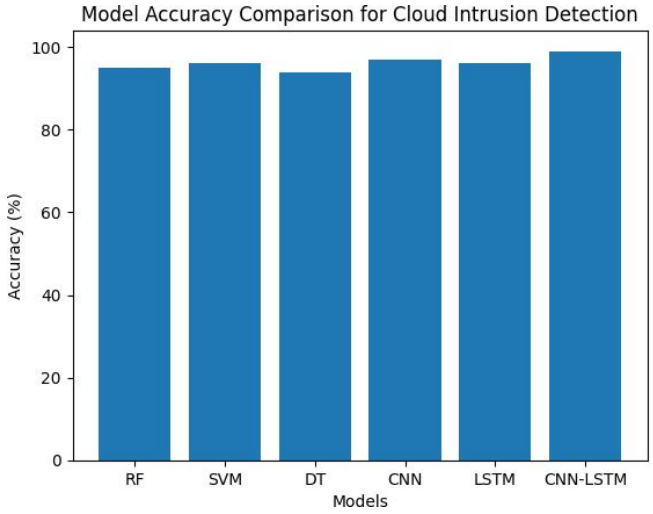


Fig. 1. Model Accuracy Comparison For Cloud Intrusion Detection

than 90Given the increasing trend of sophisticated and multi-stage cyber-attacks, deep learning-based technologies recently have gained significant attention. Convolutional Neural Net-works (CNNs) are effective to extract spatial features from net-work traffic and system metrics and therefore suitable to iden-tify volumetric attacks like DoS and DDoS attacks [5]. LSTMs have been widely used for sequential pattern recognition and thus useful to detect attacks that are persistent or sequential in nature [6]. Hybrid architectures combining CNN and LSTM models have shown improved detection capability by capturing both spatial and temporal dependencies in cloud telemetry data simultaneously [7]. Recent studies also highlight the importance of cloud-specific datasets. The ISOT-CID dataset has become a complete dataset in cloud intrusion detection studies due to its collection of telemetry data from multiple cloud layers like virtual machines, hypervisors, and network infrastructure 8. In comparison with conventional datasets like NSL-KDD or UNSW-NB15, the cloud-specific datasets more naturally represent the real cloud traffic patterns and attack behaviors, which actually help to improve model generalization and detection reliability 9. Along with the model development, deployment and lifecycle management of the machine learning models themselves have become important areas of research. For example, MLOps platforms such as MLflow and Kubeflow offer capabilities like experiment tracking, model versioning, and automated deployment workflows [10]. Further, to address the issue regarding the concept of transparency and Explainable Artificial Intelligence as well, techniques such as SHAP have also been integrated into security systems [11]. Despite all the above developments, there are still some challenges that are to be addressed. These challenges involve data imbalance issues, multitenant data privacy, adversarial attack issues related to machine learning models, and the requirement of enabling real-time processing of significant volumes of data from cloud telemetry, which is related to geographically dispersed assets [12]. The possible solution

All of the above issues can be identified using the hybrid detection aspect, which can enable the monitoring of the data in multiple layers, and the integration of the data with the cloud-native approach.

Similarly, the system presented by the authors is an extension of the research directions mentioned above by using machine learning algorithms that employ deep learning tech-niques, introducing the concept of experiment tracking and explanation techniques within the system, and in addition to that, cloud-native cloud security platforms for the deployment of the system.

III. Proposed System Architecture

The system of proposed Cloud-Native Intrusion Detection System will be observed as an end-to-end pipeline, focusing on integrating data simulation, preprocessing, model training, experiment tracking, and cloud-native deployments. The sys-tem will further be observed as modular to ensure scalability, and better integrability with existing cloud security infrastructures.

The structure starts with the data simulation layer. It integrates the creation of emulated cloud telemetry data, which includes the generation of realistic cloud operational conditions via the creation of performance data associated with the operation of virtual machines, along with logs of the operations of the hypervisor layer and network devices. Simulation also includes patterns of abnormal behaviors, including abnormal CPU peaks, suspicious memory accesses, and patterns of distributed denial-of-service traffic.

The preprocessing layer converts raw data inputs into modelable features. Preprocessing involves data cleaning, normalization, encoding, and feature scaling. For handling.

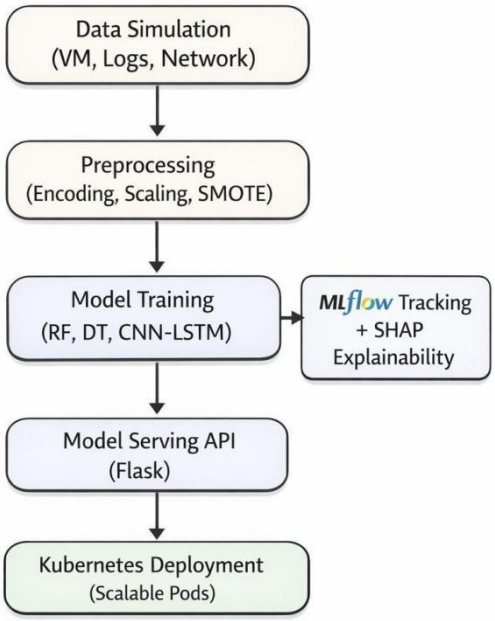


Fig. 2. Cloud-Native IDS Architecture

the inherent imbalance in security-related data, oversampling techniques are used. These include using the Synthetic Minority Over-sampling Technique. The processed data is divided into sets consisting of a training set and a test set.

The model training layer has two categories of detection models. The first category includes those traditional machine learning baseline models, such as Random Forest and Decision Tree classifiers. These models guarantee rapid inference speed and are strong for baseline detection.The second category consists of deep learning models, with a focus on a hybrid version combining a Convolutional Neural Network and Long Short-Term Memory systems. In this case, spatial relationship patterns are extracted using multi-dimensional features of telemetry data, while temporal patterns are identified with regard to attack patterns present across time series.

To ensure the reproducibility and traceability of the experiments performed, there is the inclusion of an experiment tracking layer. This is done by implementing MLOps tools. As a result, the model parameters, training metrics, dataset versions, and the results concerning the model performances, will be tracked. The modules used for the explanation of the results are based on feature attri-bution methods, which provide the security analysts with an understanding of the results.

The deployment layer follows a cloud-native microservices approach. The exposed interfaces of the trained models grant support through REST-based API service for real-time requests of prediction. The implementation of container orchestration platforms allows for automatic scaling based on the demand for traffic; therefore, having predictable and consistent perfor-mance under high load conditions. The model training layer has two categories of detection models. The first category includes those traditional machine learning baseline models, such as Random Forest and Decision Tree classifiers. These models guarantee rapid inference speed and are strong for baseline detection. The second category involves deep learning models, specifically the hybrid architecture that integrates a Convolutional Neural Network with Long Short-Term Mem-ory. The CNN component extracts spatial relationships from multi-dimensional telemetry features, while the LSTM com-ponent captures temporal attack patterns across time-series sequences.

The deployment layer follows a cloud-native microservice-based model. The exposed interfaces of the trained models provide their support through REST-based API services that can be requested in real time. An implementation of container orchestration platforms allows scaling based on the demands made on the traffic, creating predictable and consistent performance under high-load conditions. The architecture is structured in a way that can allow an update easily by re-training the models. With the architecture now, new telemetry data can be integrated into the model update process.

Overall, the proposed architecture offers the promise of a scalable, explainable, and production-ready model for IDS systems that can effectively blend the power of machine learning and cloud-native deployment options.

IV. Methodology

To begin with, the methodology that will be used to design the proposed Cloud-Native Intrusion Detection System will be based on a multi-stage pipeline, which will be responsible for transforming cloud telemetry data into actionable security intelligence. As such, the proposed workflow will cause the generation of data, the preprocessing of the proposed system, feature engineering, training, evaluation, and deployment.

A. Data Generation and Simulation

This is a synthetic data generation approach to simulate realistic cloud behavior, due to the limited availability of labeled datasets of cloud security. The generator produces multi-layer telemetry streams emulating production cloud environments, including resource utilization metrics for virtual machines like CPU, memory, and disk input-output usage, hypervisor event logs that represent VM lifecycle and system-level activities,

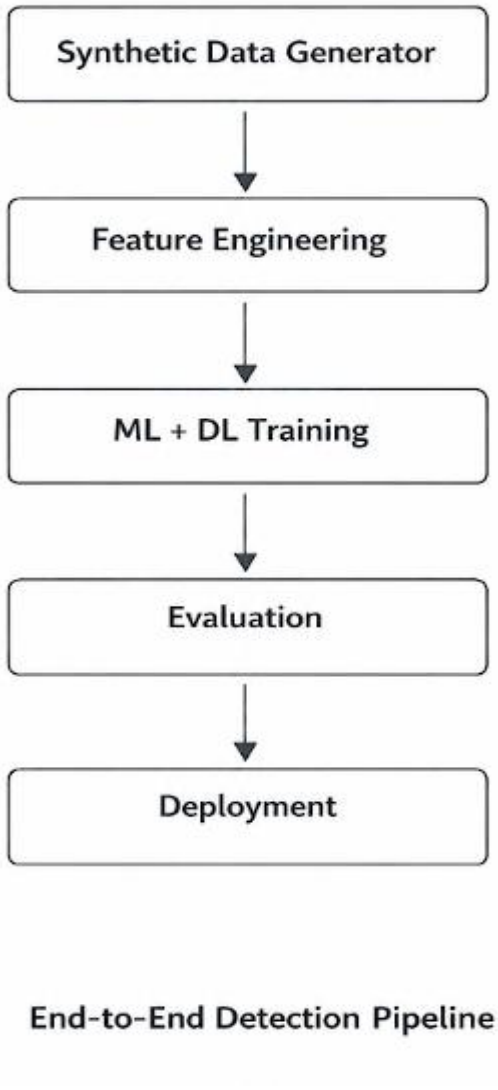


Fig. 3. End-to-End Detection pipeline

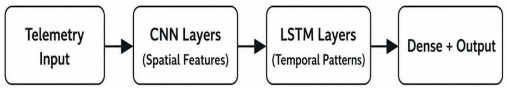
and network flow data that captures the communication pat-terns across services.

This generator will also introduce some controlled anoma-lies that best represent real attack scenarios: abnormal resource spikes, unusual log activity patterns, and network request bursts in high volumes that resemble denial-of-service behav-ior. This approach will make sure that this dataset covers the normal operational behavior and malicious activity patterns required for supervised learning.

B. *Data Preprocessing and Feature Engineering*

The preprocessing stage involves data transformation to fit the structure of the format required by the machine learning algorithms. This involves the amalgamation of data obtainable from different sources after synchronizing the timestamps. Missing values and noise are eliminated from the data.

Log attributes' categorical log types, like the types of communication protocols and event categories, are transformed into numerical forms via encoding methods. Scaling of fea-tures is used for normalizing their ranges to enable faster convergence during training.



Hybrid CNN-LSTM Detection Model

ig. 4. Cloud-Native IDS Architecture

C. *Model Training Strategy*

The detection framework combines traditional machine learning models with deep learning architectures. Baseline models such as Random Forest and Decision Tree classifiers are trained to provide fast and interpretable detection results. These models are particularly effective for structured telemetry features and provide reliable baseline performance.

Algorithm: CNN-LSTM Cloud Attack Detection
Input: Multi-modal cloud telemetry sequences
Output: Attack probability or class label

1. Collect telemetry from hypervisor, VM, and network layers
2. Perform feature normalization
3. Pass data through CNN layers
4. Extract spatial feature maps
5. Feed feature maps into LSTM layers
6. Capture sequential attack patterns
7. Pass LSTM output to fully connected layer
8. Apply Softmax activation for classification
9. Output final attack prediction

To attain an advanced level of detection, a proposed architecture based on a combination of Convolutional Neural Networks and Long Short-Term Memory is used, intended to be employed for the conduct of this research to advance the detection of computer systems from complex and multi-stage attacks.

D. *Experiment Tracking and Explainability*

In addition, experiment tracking has also been added to the training pipeline, enabling reproducibility and improvements in models. This implies that there is a recording of all configurations, and the models, with the aim of comparing different models.

It is worth mentioning that there are explainability methods that enable one to trust the predictions made by this model. Among the methods that can be used to identify what features, with regards to telemetry, have the greatest contribution in classifying this attack, is the ”feature attribution” method.

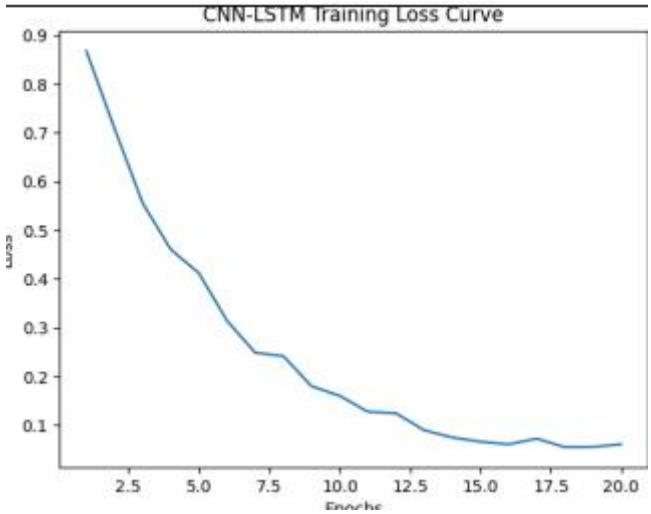


Fig. 5. CNN-LSTM Training Loss Curve

Threat Type	Detection Methods	Mitigation Strategies
Data Poisoning	Data integrity validation, anomaly pattern detection	Data sanitization, outlier removal, model retraining
Model Inversion	Differential privacy, model obfuscation	Access controls, query limiting, output perturbation
Backdoor Attacks	Neuron activation analysis, anomalous behavior detection	Model purification, retraining, input validation
Membership Inference	Privacy-preserving techniques, model generalization	Differential privacy, regularization, data augmentation

Fig. 6. Classification and Response to Adversarial Threats

E. Model Evaluation

The performance of the model is tracked and tested through multiple factors or metrics, such as the accuracy of the model, precision of the model, and the recall, F1 score, and the false positive rate. When dealing with cloud security-related task processing, it is extremely important to keep the false positive rate low to avoid overwhelming security personnel with messages or alerts.

This testing is done using unknown data for validation purposes. Intention of this testing is to validate the generality of the models. In addition to this, time-based validation strategies are used to determine if models can identify attacks in a changing cloud environment.

F. ***Deployment Preparation***

Before their deployment in the process flow, the models need to be fine-tuned to guarantee satisfactory performance in facilitating the inference process. Afterwards, they can be packaged into their container-based environment. They shall be used in a REST-based prediction service that can process real-time queries regarding threat detection. The aim of the methodology is to ensure that the system's detection capability is high while at the same time allowing for real-time processing, scalability, and reliability in the cloud.

V. IMPLEMENTATION DETAILS

The software structure of the Cloud Native Intrusion Detec-tion System is designed in such a manner that the creation, verification, and utilization of end-to-end models are enabled. It follows the concepts of reproducibility, scalability, and integration with the actual infrastructure of the cloud.

A. ***Development Environment and Technology Stack***

The development of this system is carried out through the assistance provided by a programming language called Python, depending on the robustness of machine learning, data processing, and cloud technologies. The data processing routines are coded using the assistance provided by scientific computing libraries, while the development of machine learn-ing is carried out through the assistance provided by existing machine learning and deep learning frameworks.

The experiment tracking and management of the lifecycle of ML models are facilitated by using tools that implement MLOps activities, which include logging the experiment, ver-sion control of the model, and the storage of artifacts.

B. ***Data Pipeline Implementation***

The data pipeline is driven by a training script that runs all these steps sequentially. This data pipeline starts with synthetic data generation. In this process, the generated telemetry data is created and stored in files. The files contain synthetic data that simulates data from virtual machine metrics, hypervisor events, and network flows.

The preprocessing component deals with the reading of raw telemetry data streams, cleaning, encoding, and scaling. Cleaned data is saved in optimized binary format.

Another technique used in handling class imbalance is oversampling methods. The purpose of using this technique is to ensure that all attack classes are represented by the models. After processing the data obtained from this technique, it is divided into training and test data.

C. ***Model Implementation***

Baseline models are based on ensemble and tree-based learning techniques. These models are trained with processed feature vectors and then evaluated.

The implementation of the deep learning detection model is through a hybrid model consisting of a combination of convolutional and recurrent neural networks. Automated feature extraction is done using the convolutional layers, while the sequential telemetry patterns are analyzed using the recurrent layers to train the model using the batch optimization technique with backpropagation.

Model artifacts are saved in serialized formats that can easily be loaded during the inference stage. This allows the quick deployment of the trained model without the need for retraining the model.

D. ***Experiment Tracking and Monitoring***

Experiment tracking is integrated into the training pipeline. Each training run records model parameters, dataset versions, training metrics, and output artifacts. This allows researchers and engineers to compare different model configurations and track performance improvements over time.

E. ***Code Organization and Modularity***

The implementation is kept modular to support maintain-ability and extensibility. Data generation, preprocessing, model training, evaluation, and deployment are kept as separate modules. Configuration management is implemented using a configuration file so that system parameters can easily be modified without changing the code.

This modular design facilitates testing and the development of various components independently. It is also open to integration with other external systems, and it allows for the replacement of individual components as newer techniques become available.

VI. EXPERIMENTAL RESULTS AND PERFORMANCE EVALUATION

The evaluation focuses on detection accuracy, false positive rate, and model generalization capability across simulated cloud attack scenarios.

A. ***Experimental Setup***

The experiments were carried out utilizing a dataset that is generated from the synthetic cloud telemetry simulation pipeline. The dataset comprises regular cloud operation behavior and several attack scenarios, including resource abuse anomaly and DDoS.

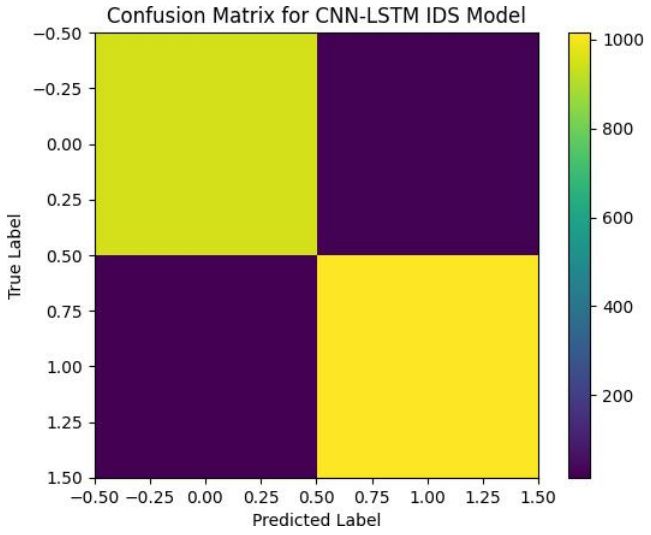


Fig. 8. Confusion Matrix For CNN-LSTM IDS Model

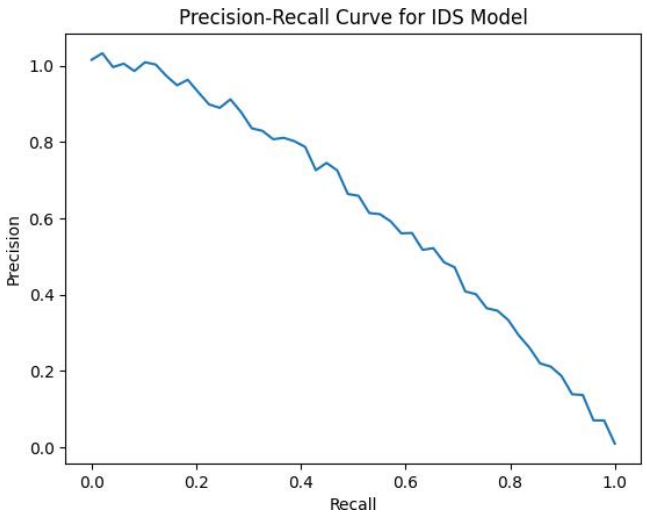


Fig. 9. Precision-Recall Curve

The data set was segmented into subsets for training and testing, which facilitated unbiased evaluation of system perfor-mance. Training data set was used for model learning, while the testing data set was set aside for detection system perfor-mance. Multiple experiments were performed to ensure stability of the results.

B. Evaluation Metrics

Intrusion detection system performance was evaluated us-ing widely accepted intrusion detection system performance metrics. Accuracy evaluates overall correctness of classifications. Precision measures the overall accuracy of detected attacks. Recall measures overall detection accuracy. The F1-score offers a balance between precision and recall. The false positive rate was measured to assess overall accuracy.

In cloud security application cases, a low false positive rate is vital because a high rate of detected alerts may confuse the security operation teams and erode faith in the whole system.

C. Baseline Machine Learning Model Performance

Traditional machine learning models showcased robust baseline detection capability. Random Forest models offered high accuracy in the detection process due to their handling capacity in high-dimensional data, particularly in the field of telemetry. Decision Tree models offered inference speed, which made them applicable to real-time systems.

The models demonstrated steady performance through a number of experiments, thereby confirming their reliability as basic detection modules in a production setting.

D. Deep Learning Model Performance

The capabilities of the hybrid CNN-LSTM model were shown in its detection power, especially for sophisticated and dynamic attack patterns. In addition, the convolutional part of the proposed model could identify spatial patterns between telemetry features, while its recurrent part could identify temporal behavior patterns.

Moreover, hybrid architecture demonstrated a superior level of recall performance compared to other baseline models, suggesting an effective detection mechanism for advanced and multi-level attack patterns. Despite high computational power requirements for deep learning models, these demonstrated improved reliability for attack detection.

E. Comparative Analysis

When comparing the results of traditional and deep learning models, baseline machine learning models offered inference speed more quickly and also with lowered computational requirements. However, the proposed hybrid deep learning model provided improved detection capability for sequential and stealth attack behaviors.

A strategy of combined detection with the use of models belonging to both categories outperformed others. Baseline models provided fast initial detection, whereas the deep learn-ing model increased the accuracy of detection for complex attack sequences.

F. Explainability and Model Interpretation Results

An explainability analysis was performed to ascertain the transparency of the model. Feature attribution tools were used to identify key features of the anomaly that contributed to the detection of the attack. Resource utilization anomalies and abnormal patterns of network traffic were the most impactful features.

It is worth mentioning that the explainability capability helped to establish confidence in the decisions taken by the machine for the detection aspect and aided the security analysts in the processes related to

G. Scalability and Deployment Performance

The prediction service deployment stability was demon-strated by the application prediction service under simulated

conditions of load. Containerized deployment of the applica-tion and the service replication ensured that the application was highly available with low latency during the prediction service response.

*H. **Discussion***

The experimental results show that integrating traditional machine learning techniques with deep learning has a positive impact on the overall detection efficacy of intrusion detection systems. The experimental results also prove the efficiency of cloud-native deployment in detecting intrusions in a distributed cloud environment.

As indicated by the evaluation process, it is clear that the suggested system is appropriate and applicable in various cloud security monitoring scenarios to solve problems.

VII. CONCLUSION AND FUTURE WORK

The current cloud environment is said to have significantly changed the deployment and management of applications, as we know it today. However, one can agree that security vulnerabilities in modern applications exist due to the dis-tributed and dynamic nature of cloud computing. It has already been established that conventional methods for dealing with intrusion detection mechanisms fail to significantly address the challenges in dealing with intrusion detection in the modern environment. To address this gap that had existed, this study introduced an intrusion detection mechanism that used machine learning and deep learning.

The proposed system includes a comprehensive flow in the implementation, which for the detection part involves generation of synthetic data, pre-processing of multi-source telemetry data that includes the baselines for the ML models, implementation of the hybrid models for the deep learning models where necessary, etc. The proposed system is designed to provide an efficient service in a real-time cloud environment with a high degree of accuracy for the detection results.

Experimental evaluation revealed that conventional machine learning models are fast and reliable for baseline detection, while hybrid deep learning models enhance detection for more sophisticated and time-dependent attack patterns. Applying explanation techniques also improves the level of transparency for the model and facilitates security teams with the analysis of detection decisions. In addition, this deployment approach offers a dynamic scaling possibility for the system regarding high-volume telemetry flows.

The implementation of a hybrid model detection strategy facilitates the rapid response and reaction to common patterns of attack while minimizing complex and emerging threats. Experimentation and model lifecycle management have also been incorporated for better improvement and reliability.

Future improvements of the system will center on the retraining of models. The long-term goal of this research is to design intelligent, adaptive, and self-improving cloud security systems that can detect and respond to cyber attacks actively, as well as ensure transparent, efficient, as well as compliant in large scale cloud environments.

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